LATE-Swarm: Late-Bound Semantic-to-Physical Commitment for Natural-Language UAV Swarm Reconfiguration

***Abstract*— Natural-language interfaces provide an intuitive way to specify swarm-level intent without requiring operators to manually define per-UAV targets, assignments, and trajectories. However, a semantically clear but under-specified formation command does not necessarily admit a unique physical realization at language-interpretation time, because its executable geometry, assignment, timing, and safety-relevant behavior may depend on swarm state and constraints that become available only later. We propose LATE-Swarm, a command-to-control framework for late-bound, authority-bounded task execution in UAV swarms. LATE-Swarm preserves unresolved execution-relevant intent in a Candidate Language-to-Formation Specification and progressively commits numerical decisions as the required state, geometry, assignment, and feasibility information becomes available. It further separates predictable nominal inter-agent risk from residual execution-time risk through safety-aware assignment and feedback correction, while constraining motion-style effects to admissible execution choices within the resolved safety and feasibility envelope. Multi-UAV simulation experiments support the proposed commitment strategy and planning–execution safety decomposition, while demonstrating end-to-end execution across under-specified and compositional formation missions.**

I. INTRODUCTION

Specifying multi-robot coordination becomes increasingly demanding as team size and task coupling grow. For unmanned aerial vehicle (UAV) swarms, natural language lets operators express formation intent without manually defining per-UAV targets, assignments, and trajectories. Formation control is well established [1], and large language models (LLMs) have translated language instructions into structured plans and executable robot behaviors [2], [3]. We study natural-language-driven UAV-swarm formation reconfiguration, where collective semantic intent must be realized as coordinated physical motion.

A key difficulty is that a semantically clear but under-specified formation command may require different numerical realizations across swarm states. In Fig. 1, the same command defines the same target circle for two swarms with an identical centroid, but their different initial configurations induce different UAV-to-slot travel distances after assignment. Because automatic timing is resolved from assignment-dependent travel distance under fixed motion limits, the resulting execution durations differ. Thus, a semantically well-defined formation command does not necessarily determine a unique executable physical realization at interpretation time.

Recent language-grounded multi-robot systems have used LLMs for task decomposition and allocation [4], multi-robot communication and coordination [5], [6], and feedback-driven plan revision [7]. Other approaches combine language reasoning with structured optimization [8], classical

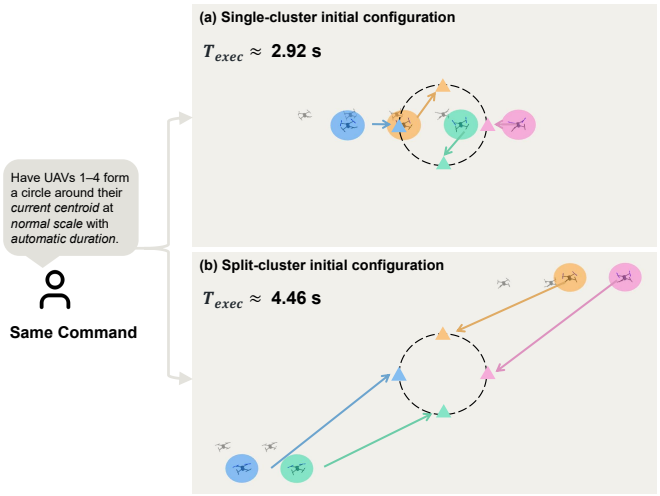


Fig. 1. Initial configurations and automatic execution durations for a shared formation command and centroid.

planning [9], [10], scheduling [11], or behavior-tree-guided code generation [12]. In UAV and swarm settings, natural-language instructions have further been mapped to geometric formations [13], coordinated trajectories [14], and language-generated motion references with safety-constrained refinement [15]. Structured multi-UAV commands have also been coupled with state-aware supervision [16], while LLM-guided distributed predictive control has been studied for decentralized UAV formations [17].

Closer to the semantic-to-physical realization boundary, robotic decisions are grounded through skill affordances [18], multimodal state observations [19], and spatial value maps [20]. State-aware grounding methods use environment representations and feedback to update or revise language-derived decisions [21]–[23], while task–motion approaches couple semantic goals or continuous action parameters with geometric reasoning, formal planning, and motion-feasibility checks [24]–[27]. Execution monitoring addresses plan–execution mismatch [28]; conformal prediction quantifies planning uncertainty under successful-execution assumptions [29]; and safety-aware language planning incorporates safety feedback and control constraints [30]. Separately, classical safety barrier certificates modify nominal controllers through constrained optimization [31]. In under-specified UAV-swarm formation reconfiguration, however, semantic interpretation can be complete before some execution-relevant quantities are uniquely determined: a relative center depends on the activated swarm state, formation scale may depend on geometry and safety-related spacing, and automatic duration may depend on assignment-dependent

travel distance. This leaves a specific commitment problem: how to represent such physical quantities while they remain under-determined, and how to bind each of them only when the state, geometry, assignment, or feasibility information required to determine it becomes available.

To address these challenges, we propose **LATE-Swarm**, a framework for **L**ate-bound, **A**uthority-bounded **T**ask **E**xecution in UAV swarms. Its design follows an information-aligned, authority-bounded commitment principle: each execution-relevant decision is committed only when the information required to resolve it becomes available, while each stage may modify only variables within its predefined authority. An LLM translates language commands into a Candidate Language-to-Formation Specification (Candidate LFS), preserving unresolved requests for downstream realization. Using the current swarm state, the framework resolves formation geometry and target slots, then determines a safety-aware UAV-to-slot assignment before committing the execution duration from assignment-dependent travel distances and motion limits. This assignment reduces predictable conflicts in nominal motion, while closed-loop feedback mitigates residual proximity risks revealed during execution. Across planning and execution, motion style acts only on delegated execution variables, affecting automatic timing and designated tracking parameters after safety- and feasibility-critical quantities have been resolved.

The contributions of this paper are as follows:

- **Late-bound semantic-to-physical commitment with bounded authority.** We develop LATE-Swarm, a command-to-control framework that preserves unresolved execution-relevant intent in a Candidate LFS and commits physical decisions only as the required state, geometry, assignment, and feasibility information becomes available. The same framework gives motion style bounded authority over automatic timing and designated tracking parameters, while keeping the hard-risk policy and feasible explicit timing commitments outside that authority.
- **Planning–execution safety decomposition for swarm reconfiguration.** We develop a planning–execution safety decomposition that assigns predictable nominal inter-agent conflicts to planning and residual risks revealed during physical realization to feedback, thereby aligning safety responsibility with information availability.

II. Problem Formulation

We consider a swarm of N UAVs, indexed by $\mathcal{N} = \{1, \dots, N\}$, performing formation reconfiguration in a three-dimensional workspace. The state of UAV i at time t is denoted by $\mathbf{x}_i(t) = [\mathbf{p}_i^\top(t), \mathbf{v}_i^\top(t)]^\top$, where $\mathbf{p}_i(t), \mathbf{v}_i(t) \in \mathbb{R}^3$ denote its position and velocity, respectively. At the start of state-dependent task realization, the observable swarm state is $X_0 = \{\mathbf{x}_i(t_0)\}_{i\in\mathcal{N}}$. Given a natural-language command $\ell \in \mathcal{L}$ and X_0, the problem is to determine a task-level executable realization Θ, whose closed-loop execution from X_0 induces a realized swarm behavior Ξ. We therefore distinguish $(\ell, X_0) \mapsto \Theta$ and $(\Theta, X_0) \xrightarrow{\text{closed-loop execution}} \Xi$.

We formulate a single formation-reconfiguration task below. Sequential missions compose such tasks over time, while parallel groups resolve their member tasks from a shared swarm-state snapshot. At the problem level, the task-relevant semantics expressed by ℓ are abstracted as $\sigma_\ell = (\mathcal{N}_\ell, f, \tilde{\mathbf{c}}, \tilde{\rho}, \tilde{T}, m, s)$, where $\mathcal{N}_\ell \subseteq \mathcal{N}$ is the participating UAV set, $f \in \mathcal{F}$ is the requested formation type, $\tilde{\mathbf{c}}$, $\tilde{\rho}$, and $\tilde{T}$ denote semantic specifications of formation center, scale, and execution duration, respectively, and $m \in \mathcal{M}$ and $s \in \mathcal{S}$ denote motion-style and safety preferences. A tilde indicates a language-level semantic specification that need not yet correspond to a unique numerical physical value.

The task-level executable realization is defined as $\Theta = (\mathbf{c}^\star, \rho^\star, \mathcal{P}^\star, \pi^\star, T_{\text{exec}}, m, s)$, where $\mathbf{c}^\star \in \mathbb{R}^3$ and $\rho^\star > 0$ denote the resolved numerical formation center and scale. Let $n_\ell = |\mathcal{N}_\ell|$. The set $\mathcal{P}^\star = \{\mathbf{p}_j^\star\}_{j=1}^{n_\ell}$ contains the target formation slots induced by f, $\mathbf{c}^\star$, $\rho^\star$, and n_ℓ, while $\pi^\star : \mathcal{N}_\ell \to \{1, \dots, n_\ell\}$ is a bijective assignment from participating UAVs to these slots. $T_{\text{exec}} > 0$ denotes the resolved numerical execution duration.

Executing Θ from X_0 induces the closed-loop swarm behavior $\Xi = \{\xi_i\}_{i\in\mathcal{N}}$, where $\xi_i = \{\mathbf{x}_i(t), \mathbf{u}_i(t)\}_{t_0 \le t \le t_f}$, $\mathbf{u}_i(t)$ denotes the acceleration command sent to UAV i, and t_f denotes the task termination time. The realized inter-agent separation is $d_{ij}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2$. Given a prescribed hard-risk proximity threshold $d_{\text{hard}} > 0$, a hard-risk proximity event occurs when $d_{ij}(t) < d_{\text{hard}}$. The objective is therefore to determine Θ and realize Ξ while respecting the field-specific semantics of explicit requests, the applicable physical constraints, and the predefined authority of semantic preferences, while mitigating hard-risk events. Motion style m and safety preference s have distinct authority: s may affect safety-related admissibility margins, whereas m may affect only delegated timing and tracking behavior and cannot override resolved safety parameters, motion limits, or a dynamically feasible explicit duration.

III. Method

LATE-Swarm realizes under-specified formation commands through five modules (Fig. 2): *(i)* Semantic Specification, *(ii)* State-Aware Formation Generator, *(iii)* Safety-Aware Assignment, *(iv)* Motion Generator, and *(v)* Feedback Safety and Tracking.

A. Semantic Specification

Given a natural-language command ℓ, the Semantic Specification module uses an LLM to encode the problem-level task semantics into a structured Candidate Language-to-Formation Specification (Candidate LFS),

$$\tau^c = (\mathcal{N}_\ell, f, \tilde{\mathbf{c}}, \tilde{\rho}, \tilde{T}, m, s, q),$$

where $(\mathcal{N}_\ell, f, \tilde{\mathbf{c}}, \tilde{\rho}, \tilde{T}, m, s)$ retain the semantics defined in Sec. II, and q denotes task-transition semantics for mission composition.

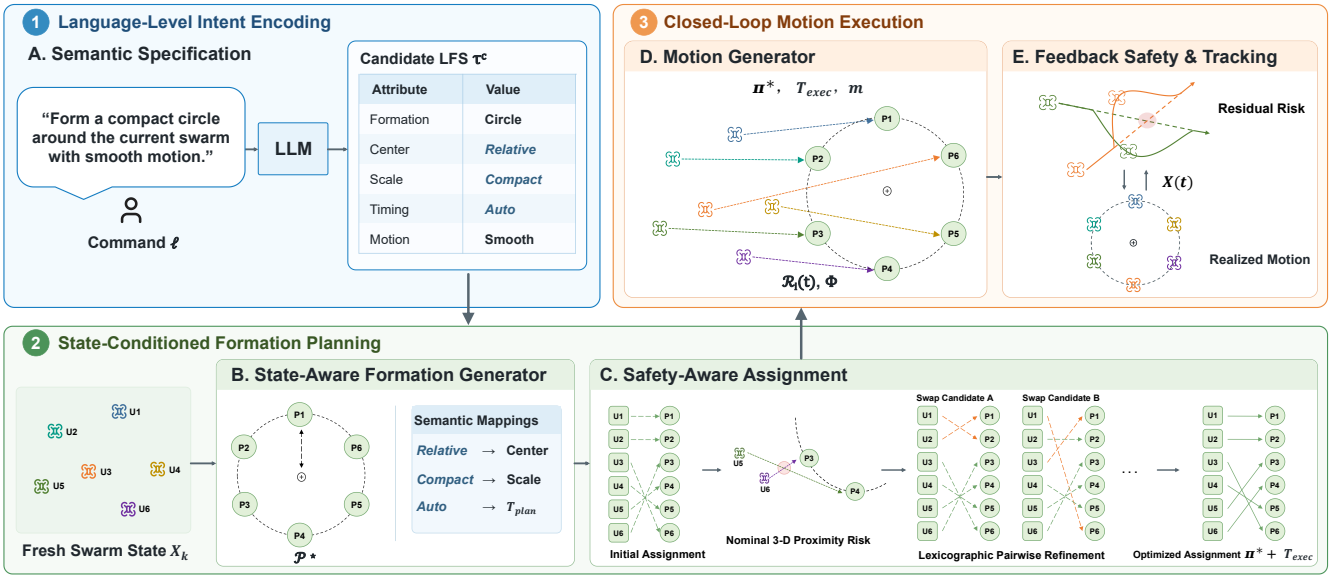


Fig. 2. Overview of the LATE-Swarm framework. The Semantic Specification module converts a natural-language command into a Candidate LFS. The State-Aware Formation Generator resolves the formation geometry and planning duration. The Safety-Aware Assignment module determines the UAV-to-slot assignment and final execution duration. The Motion Generator converts the executable task into per-UAV nominal motion. Finally, the Feedback Safety and Tracking module mitigates residual proximity risks during closed-loop execution and realizes the commanded swarm motion.

The semantic values of $\mathcal{N}_\ell$, f, m, s, and q can be determined without later geometric or assignment information. In contrast, $\tilde{\mathbf{c}}$, $\tilde{\rho}$, and $\tilde{T}$ may contain explicit numerical requests or remain relative, qualitative, or automatic requests. Explicit numerical requests are retained with field-specific semantics and evaluated by deterministic downstream admissibility rules, while unresolved fields are committed only when their required runtime information becomes available. After schema and state-independent semantic checks, a valid Candidate LFS proceeds to state-conditioned realization without requiring all fields to be numerical.

B. State-Aware Formation Generator

Once a validated Candidate LFS is activated, the State-Aware Formation Generator obtains a fresh swarm-state snapshot $X_k = \{\mathbf{x}_i(t_k)\}_{i\in\mathcal{N}_\ell}$ of the participating UAVs with $X_k = X_0|_{\mathcal{N}_\ell}$ for the single-task formulation. The module resolves the numerical formation center $\mathbf{c}^\star$ and scale $\rho^\star$, generates the target-slot set $\mathcal{P}^\star$, and computes an assignment-independent provisional planning duration T_{plan} from a conservative travel-distance bound.

Let $\bar{\mathbf{p}}_k = n_\ell^{-1}\sum_{i\in\mathcal{N}_\ell}\mathbf{p}_i(t_k)$ denote the current centroid of the participating UAVs. An explicitly specified center is preserved; a relative request applies its world-frame offset to $\bar{\mathbf{p}}_k$; and an automatic or current-centroid request resolves to $\bar{\mathbf{p}}_k$. Formation f defines a deterministic unit geometry $U_f(n_\ell) = \{\mathbf{g}_j^f\}_{j=1}^{n_\ell}$ with minimum unit-slot spacing

$$\delta_f(n_\ell) = \min_{j\neq l}\|\mathbf{g}_j^f - \mathbf{g}_l^f\|_2.$$

The scale request is mapped to a numerical requested scale ρ_{req}: an explicit value is used directly as the requested scale, qualitative labels apply fixed calibrated multipliers to a nominal scale, and an automatic request uses that nominal scale. The safety preference s is compiled into a planning separation $d_{\text{plan}}(s) \geq d_{\text{hard}}$, while d_{hard} itself remains the fixed hard-risk threshold. The final scale and slots are

$$\rho^\star = \max\left\{\rho_{\text{req}}, \frac{d_{\text{plan}}(s)}{\delta_f(n_\ell)}\right\}, \qquad \mathbf{p}_j^\star = \mathbf{c}^\star + \rho^\star\mathbf{g}_j^f.$$

An explicit scale is preserved when the resulting slot spacing meets $d_{\text{plan}}(s)$; otherwise, it is increased to the minimum admissible value. If this geometry cannot satisfy the workspace bounds, the task is rejected.

We compute the conservative assignment-independent distance bound

$$D_{\text{plan}} = \max_{\substack{i\in\mathcal{N}_\ell \\ j\in\{1,\dots,n_\ell\}}} \|\mathbf{p}_i(t_k) - \mathbf{p}_j^\star\|_2,$$

which provides a pre-assignment bound for provisional timing and dynamic-feasibility evaluation. For the zero-endpoint-derivative Minimum-Jerk reference [32] used downstream, the minimum reference-feasible duration for displacement D is

$$T_{\min}(D) = \max\left\{T_0, \frac{1.875D}{v_{\max}}, \sqrt{\frac{10D}{\sqrt{3}a_{\max}}}, \left(\frac{60D}{j_{\max}}\right)^{1/3}\right\},$$

where T_0 is the minimum allowed duration and $v_{\max}$, $a_{\max}$, and $j_{\max}$ are the configured motion limits. Accordingly,

$$T_{\text{plan}} = \begin{cases} \max\{T_{\text{req}}, T_{\min}(D_{\text{plan}})\}, & \tilde{T} \text{ is explicit}, \\ \alpha_T(m)T_{\min}(D_{\text{plan}}), & \tilde{T} \text{ is automatic}, \end{cases}$$

where T_{req} is the explicit requested duration and $\alpha_T(m) \geq 1$ is the motion-style factor used only for automatic timing. T_{plan} is a pre-assignment reference-feasibility estimate; the final duration is recomputed from assignment-dependent travel distances.

C. Safety-Aware Assignment

Given the target-slot set $\mathcal{P}^\star$ and current UAV positions in X_k, the Safety-Aware Assignment module determines a UAV-to-slot mapping while accounting for nominal inter-agent proximity and travel efficiency. Robot-to-goal assignment is related to multi-robot task allocation [33], and its coupling with trajectory planning has been studied in CAPT [34]. We first compute the distance-minimizing assignment

$$\pi_0 = \arg\min_{\pi} \sum_{i\in\mathcal{N}_\ell} \left\| \mathbf{p}_i(t_k) - \mathbf{p}^\star_{\pi(i)} \right\|_2,$$

using the Hungarian algorithm [35].

Candidate assignments are evaluated using a synchronized equal-progress nominal surrogate from the planning snapshot X_k. Let

$$\beta(\lambda) = 10\lambda^3 - 15\lambda^4 + 6\lambda^5, \qquad \lambda \in [0,1].$$

For assignment π, UAV i follows the nominal equal-progress path

$$\mathbf{r}_i^\pi(\lambda) = \mathbf{p}_i(t_k) + \beta(\lambda)\left(\mathbf{p}^\star_{\pi(i)} - \mathbf{p}_i(t_k)\right).$$

The predicted pairwise closest approach is

$$d_{ij}^\pi = \min_{\lambda\in[0,1]} \left\| \mathbf{r}_i^\pi(\lambda) - \mathbf{r}_j^\pi(\lambda) \right\|_2,$$

which is evaluated analytically in three dimensions for these equal-progress trajectories. Because all UAVs share the same normalized progress, this spatial closest-approach metric is invariant to a common time scaling and therefore does not depend on T_{plan} for a single synchronized task. This invariance allows assignment to be resolved before the final duration is committed, avoiding a circular dependence between proximity evaluation and timing.

Assignments are compared using

$$C(\pi) = \left(N_{\text{hard}}(\pi), J_{\text{margin}}(\pi), J_{\text{distance}}(\pi)\right),$$

where

$$N_{\text{hard}}(\pi) = \sum_{i<j} \mathbb{I}[d_{ij}^\pi < d_{\text{hard}}],$$

$$J_{\text{margin}}(\pi) = \sum_{i<j} \left[\max\left(0, \frac{d_{\text{plan}}(s) - d_{ij}^\pi}{d_{\text{plan}}(s)}\right)\right]^2,$$

$$J_{\text{distance}}(\pi) = \sum_{i\in\mathcal{N}_\ell} \left\| \mathbf{p}_i(t_k) - \mathbf{p}^\star_{\pi(i)} \right\|_2.$$

Here, N_{hard} counts violations of the hard-risk threshold d_{hard}, whereas J_{margin} penalizes intrusion into the preferred planning margin $d_{\text{plan}}(s)$; J_{distance} measures total travel distance. The lexicographic order first minimizes nominal hard-risk violations, followed by planning-margin intrusion and travel distance.

Candidate mappings are generated by pairwise exchanges of target slots from π_0. A swap is accepted whenever it strictly improves $C(\pi)$ lexicographically, and refinement stops when no pairwise exchange improves the objective, yielding the final assignment $\pi^\star$.

Once $\pi^\star$ is fixed, the assignment-dependent travel distances are

$$D_i^\star = \left\| \mathbf{p}_i(t_k) - \mathbf{p}^\star_{\pi^\star(i)} \right\|_2, \qquad D_{\text{exec}} = \max_i D_i^\star.$$

Applying the same timing policy as in the preceding module to D_{exec} yields the final duration T_{exec}. Since $D_{\text{exec}} \le D_{\text{plan}}$ and $T_{\min}(D)$ is nondecreasing, this step can shorten the provisional duration while retaining the reference-feasibility bound for the selected assignment. A final nominal separation below d_{hard} rejects the task, while d_{plan} intrusion is retained as a nominal margin violation. At this point, the resulting resolved quantities instantiate the task-level executable realization Θ.

For a parallel group, all member tasks are resolved from a common swarm-state snapshot. The group planning separation is set to the most conservative member value, $d_{\text{plan}}^G = \max_g d_{\text{plan}}^{(g)}$. UAV-to-slot assignments remain restricted to their respective member-task groups, while nominal inter-UAV proximity is evaluated jointly over all participating UAVs. The same lexicographic assignment objective is then applied to the grouped motion. For synchronized completion, each member task is first resolved to its assignment-dependent duration $T_{\text{exec}}^{(g)}$, after which the group duration is committed as $T_G = \max_g T_{\text{exec}}^{(g)}$.

D. Motion Generator

Once the optimized assignment $\pi^\star$ and final execution duration T_{exec} are fixed, the Motion Generator converts the committed task into per-UAV execution specifications. Upon command acceptance, UAV i initializes its local Minimum-Jerk reference [32] from its latest measured position to the assigned target $p^\star_{\pi^\star(i)}$, using the committed duration T_{exec}, and is represented as

$$\mathcal{R}_i(t) = \left(\mathbf{p}_i^{\text{nom}}(t), \mathbf{v}_i^{\text{nom}}(t), \mathbf{a}_i^{\text{nom}}(t)\right), \quad t \in [t_k, t_k + T_{\text{exec}}].$$

The module also compiles the task-level Execution Profile

$$\Phi = \left(T_{\text{exec}}, \boldsymbol{\omega}_c, \boldsymbol{\omega}_o, \mathcal{L}_{\text{dyn}}, \mathcal{S}_{\text{exec}}\right),$$

where $\mathcal{L}_{\text{dyn}} = (v_{\max}, a_{\max}, j_{\max})$ contains the configured motion limits and $\mathcal{S}_{\text{exec}} = (d_{\text{enter}}(s), d_{\text{exit}}(s), k_{\text{rep}}(s))$ contains the resolved soft safety parameters used during execution.

With T_{exec}, motion limits, and resolved safety parameters fixed, motion style m modifies only the designated tracking parameters through a bounded gain $\kappa(m)$,

$$\boldsymbol{\omega}_c = \kappa(m)\bar{\boldsymbol{\omega}}_c, \qquad \boldsymbol{\omega}_o = \kappa(m)\bar{\boldsymbol{\omega}}_o,$$

where $\bar{\boldsymbol{\omega}}_c$ and $\bar{\boldsymbol{\omega}}_o$ are the baseline control and observer bandwidth vectors used by the downstream linear active disturbance rejection controller (LADRC).

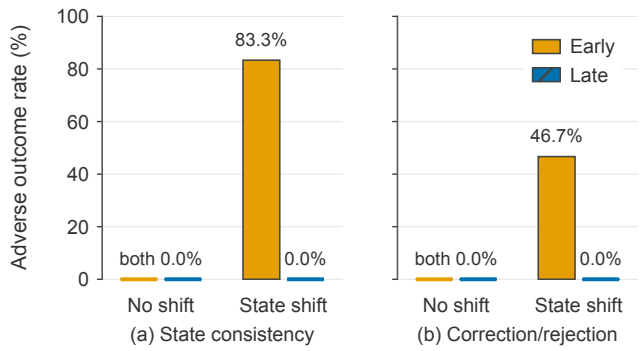


Fig. 3. Early and Late commitment under unchanged and shifted execution states: (a) state-consistency violations; (b) feasibility correction/rejection.

E. Feedback Safety and Tracking

Planning-to-dispatch state evolution, command-start skew, tracking errors, disturbances, and evolving inter-UAV states can introduce residual proximity risk not captured by nominal assignment-time prediction. The Feedback Safety and Tracking module therefore uses the nominal reference $\mathcal{R}_i(t)$, Execution Profile Φ, and current measured swarm state $X(t) = \{\mathbf{x}_i(t)\}_{i\in\mathcal{N}}$ to apply IAPF corrections and closed-loop tracking. All UAVs in $\mathcal{N}$, including nonparticipants, are considered as dynamic neighbors.

For UAVs i and j, we define the closing speed

$$\nu_{ij}(t) = -\frac{(\mathbf{p}_i(t) - \mathbf{p}_j(t))^\top (\mathbf{v}_i(t) - \mathbf{v}_j(t))}{d_{ij}(t)},$$

where $\nu_{ij} > 0$ indicates that the pair is approaching. For an inactive pair, avoidance is triggered when

$$d_{ij} < d_{\text{enter}}(s) \quad \text{and} \quad (\nu_{ij} > 0 \text{ or } d_{ij} < d_{\text{hard}}).$$

This velocity-aware condition suppresses activation for separating pairs outside the hard-risk region. Once activated, the pair remains active until $d_{ij} \geq d_{\text{exit}}(s)$, with $d_{\text{hard}} < d_{\text{enter}}(s) < d_{\text{exit}}(s)$, introducing hysteresis around the activation boundary.

For active neighbors, a standard distance-dependent repulsive term [36], scaled by $k_{\text{rep}}(s)$, is accumulated over the active neighbor set. For a receding pair outside the hard-risk region, the repulsion is attenuated as the pair approaches the exit boundary. A deterministic three-dimensional escape component is used to break symmetry in geometrically ambiguous encounters. The accumulated action is converted into bounded and temporally filtered position and acceleration corrections, yielding

$$\mathbf{p}_i^{\text{safe}}(t) = \mathbf{p}_i^{\text{nom}}(t) + \Delta\mathbf{p}_i^{\text{IAPF}}(t),$$
$$\mathbf{a}_i^{\text{safe}}(t) = \mathbf{a}_i^{\text{nom}}(t) + \Delta\mathbf{a}_i^{\text{IAPF}}(t).$$

The corrected reference is tracked by LADRC [37], [38], using the bandwidths (ω_c, ω_o) in Φ, generating the acceleration commands sent to the UAV autopilot.

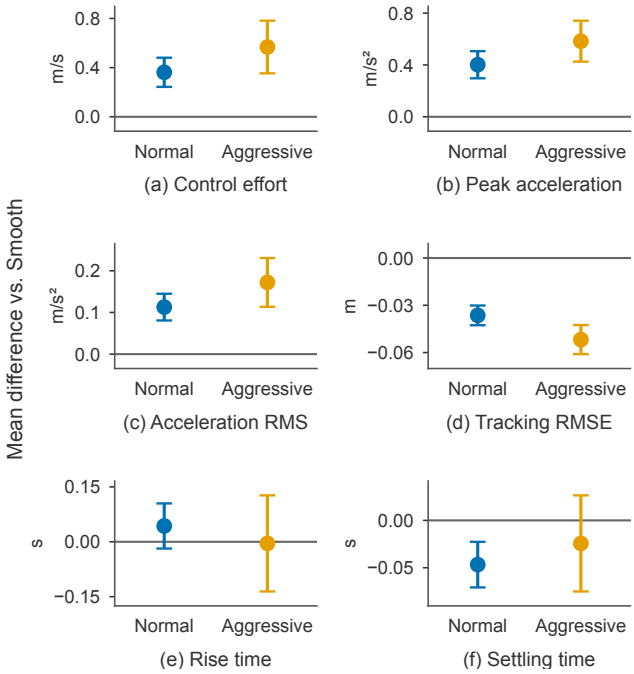


Fig. 4. Paired motion-style effects relative to Smooth under matched scenarios. Markers show mean paired differences with two-sided 95% Student-t confidence intervals.

TABLE I
MOTION-STYLE AUTHORITY CHECKS

Case	Request / Bound	Value (S/N/A)	Pres.
Feas. exp. T	$4.00 > 2.884$ s	4.00 / 4.00 / 4.00	15/15
Infeas. exp. T	$1.50 < 2.884$ s	2.884 / 2.884 / 2.884	15/15
Auto T	$T_{\min} = 2.884$ s	3.750 / 3.317 / 3.173	15/15
Safety-active	$d_{\text{hard}} = 1.50$ m	1.50 / 1.50 / 1.50	15/15
Overall	–	0 overrides	60/60

Note: S/N/A: Smooth/Normal/Aggressive. Values are T_{exec} for timing cases and d_{hard} for safety-active execution. Pres. counts attempts satisfying all applicable authority predicates with zero unauthorized overrides.

IV. EXPERIMENTS

A. Late-Bound Semantic-to-Physical Commitment

Preserving Under-Specified Semantics: We evaluate Candidate LFS on a preregistered valid-command set spanning relative or automatic center specifications, qualitative or automatic scale requests, explicit or automatic timing, and motion-style semantics. We measure exact semantic-task fidelity together with premature numerical commitment in fields that should remain unresolved at interpretation time. Across 96 valid commands containing 225 registered unresolved-value instances, Candidate LFS achieves 100% exact semantic-task accuracy and no premature numerical commitments, showing that task-relevant intent is preserved without inventing state-dependent physical values.

Commitment Timing Under State Shift: To examine when delayed commitment matters for physical realization, we compare Early and Late commitment in a controlled paired experiment over six preregistered semantic families, with ten independently sampled physical-state realizations per family. For each of the resulting 60 state units, the two strategies

TABLE II

PLANNING AND REALIZED SAFETY OUTCOMES BY FACTORIAL CONDITION

Experimental condition		Planning	Execution		
Risk family	Condition	Pred. conflicts	J_{hard} [pair-s]	$d_{\min}$ [m]	Any risk [%]
Predictable $N = 23$	P_0F_0	2.0	1.738	0.228	100.0
	P_0F_1	2.0	1.624	0.250	100.0
	P_1F_0	0.0	0.000	2.068	0.0
	P_1F_1	0.0	0.000	2.068	0.0
Residual $N = 26$	P_0F_0	0.0	1.661	1.400	100.0
	P_0F_1	0.0	0.305	1.468	57.7
	P_1F_0	0.0	1.488	1.428	88.5
	P_1F_1	0.0	0.389	1.459	61.5
Mixed $N = 25$	P_0F_0	2.0	3.242	0.525	100.0
	P_0F_1	2.0	1.880	0.624	100.0
	P_1F_0	0.0	1.369	1.420	88.0
	P_1F_1	0.0	0.330	1.462	52.0

Note: P_1/F_1: planning/feedback enabled. Pred. conflicts: mean predicted hard-conflict count; J_{hard}: realized hard-risk exposure (pair-s); $d_{\min}$: mean realized minimum inter-UAV separation; Any risk: fraction of complete blocks with at least one realized hard-risk event. N: complete four-condition blocks per risk family.

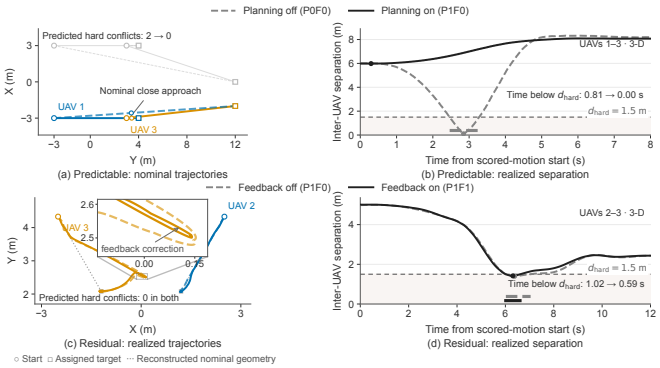


Fig. 5. Representative trajectory and 3-D separation comparisons: (a,b) planning off/on for Predictable risk; (c,d) feedback off/on for Residual risk. Bottom segments mark intervals below the hard-risk threshold d_{hard}.

share the same command semantics, Candidate LFS, parse state, execution state, resolution policy, and downstream allocation, and differ only in the stage at which unresolved numerical values are committed. Each unit is evaluated under both No shift, where the execution-relevant state remains unchanged after interpretation, and State shift, where it changes before execution. Both strategies are evaluated against an independent execution-state oracle that derives the expected center, scale, and assignment-dependent timing from the Candidate semantics, execution state, and frozen policy. We measure state-consistency violations and downstream feasibility correction or rejection.

Fig. 3 shows a state-dependent separation between the two commitment strategies. Under No shift, neither strategy produces a state-consistency violation or a feasibility correction/rejection in any of the 60 paired state units. Under State shift, Early commitment violates execution-state consistency in 50/60 units and requires downstream feasibility correction in 28/60, whereas Late commitment produces neither outcome in any of the 60 units. The corresponding paired rate differences are 83.3 percentage points for state-consistency violations and 46.7 percentage points for correction/rejection. Because the two strategies remain equivalent under No shift, the observed separation is specific to changes in execution-relevant state after interpretation.

B. Bounded Motion-Style Semantics

We evaluate whether motion-style semantics produce measurable closed-loop behavioral changes while remaining subordinate to resolved safety parameters, reference-feasible timing, and feasible explicit timing requirements.

Behavioral Effects of Motion Style: Motion-style effects are isolated using 15 paired scenario–seed triplets, each executed with smooth, normal, and aggressive profiles. The initial state, assigned targets, explicit duration, and Minimum-Jerk nominal reference are held fixed, isolating the tracking-parameter channel of motion style. The authority checks below examine automatic timing. We measure actuation intensity through control effort $\int \|\mathbf{a}_{\mathrm{cmd}}\| dt$ and peak/RMS acceleration, tracking accuracy through RMSE, and transient response through 10–90% acceleration rise time and swarm settling time. Fig. 4 reports the corresponding paired effects relative to Smooth.

From Smooth through Normal to Aggressive, mean control effort and peak/RMS acceleration increase, while tracking RMSE decreases, indicating stronger control action and tighter tracking. Rise-time effects remain near zero, and settling time varies nonmonotonically across styles.

Authority Bounds of Motion Style: We next evaluate four registered cases covering feasible explicit timing, infeasible explicit timing, automatic timing, and safety-active execution. The primary metric is the Priority-Preservation Rate, i.e., the fraction of attempts satisfying all applicable authority predicates without unauthorized overrides; Table I reports the corresponding style-wise outcomes.

All 60 registered attempts satisfy the authority predicates with zero unauthorized overrides. Across the four cases, feasible explicit timing remains style-invariant, automatic timing changes with style only above the feasibility bound,

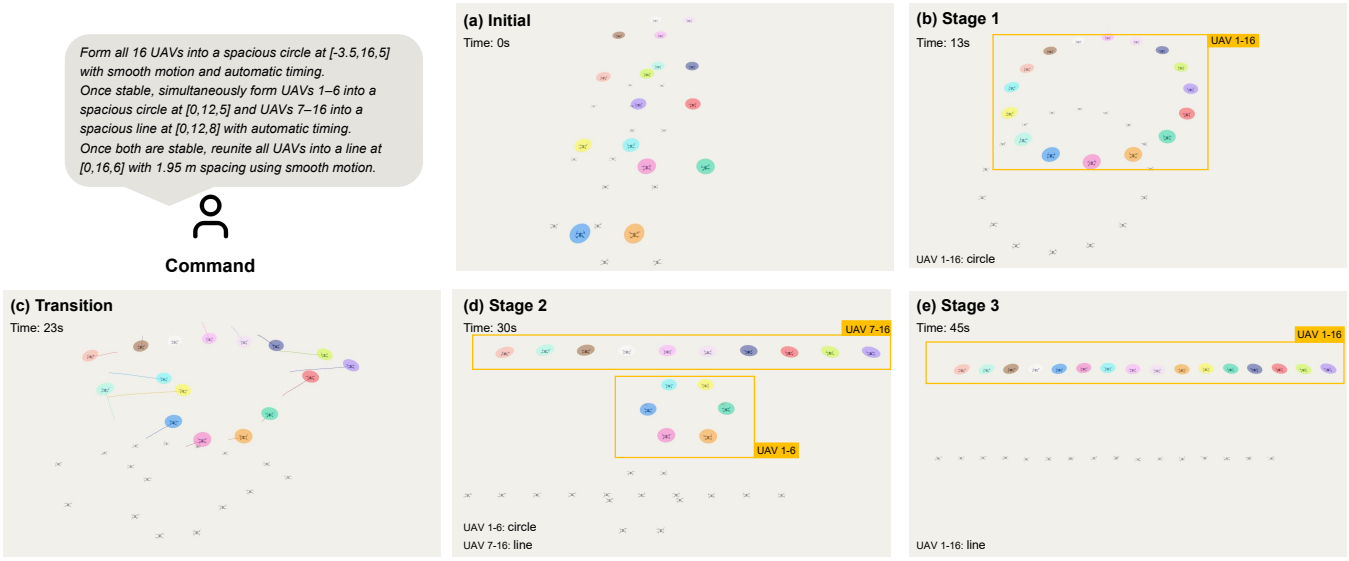


Fig. 6. Three-stage compositional mission with 16 UAVs in Gazebo: (a) initial state; (b,d,e) completed stages; (c) transition to Stage 2. Colored trails show recent UAV motion and direction; timestamps give elapsed mission time in seconds.

infeasible timing is constrained by feasibility independently of style, and the fixed hard-risk threshold remains unchanged.

C. Planning–Execution Safety Decomposition

We evaluate the planning–execution safety decomposition using three preregistered risk families: Predictable, Residual, and Mixed, under a $P_0/P_1 \times F_0/F_1$ factorial design, where P_1 enables safety-aware planning and F_1 enables feedback safety. Fig. 5 illustrates representative mechanisms, and Table II reports the population-level outcomes. The primary metric is realized hard-risk exposure J_{hard}. Planning and feedback main effects average the respective on/off contrasts over the other factor. The Mixed-family comparison additionally measures the feedback effect with planning enabled.

Predictable Risk: Planning reduces J_{hard} by 1.681 pair-s (95% CI: 1.469–1.979 in magnitude) and removes the registered predicted hard conflicts, whereas the feedback main effect is near zero. Thus, conflicts already visible in nominal motion are primarily addressed at the planning stage.

Residual Risk: All conditions contain zero predicted hard conflicts, and the observed planning main effect is near zero. Feedback reduces J_{hard} by 1.228 pair-s (95% CI: 1.056–1.389 in magnitude) and lowers the occurrence of realized hard-risk events, supporting execution-time mitigation of risk not captured by nominal planning.

Mixed Risk: When both risk sources coexist, planning removes the registered nominal hard conflicts, while enabling feedback after planning further reduces J_{hard} by 1.038 pair-s. These results support complementary planning and feedback roles in the tested domain. Nonzero event rates in the Residual and Mixed families show that threshold crossings persist with feedback; together, event rates and cumulative exposure characterize risk occurrence and extent.

D. End-to-End Swarm Evaluation

The complete LATE-Swarm command-to-control pipeline is evaluated in a separately preregistered end-to-end campaign. The evaluation includes prospectively feasible under-specified commands with relative or maintain-current centers, qualitative scales, and automatic timing, together with SIMPLE, UNDER-SPECIFIED, and COMPOSITIONAL missions at $N = 8$, 12, and 16. A representative closed-loop execution in Gazebo [39] is shown in Fig. 6. Each sequential task is resolved from the current swarm state, carrying the preceding stage's outcome into the next request. Parallel tasks share a snapshot and undergo joint nominal proximity evaluation.

All 60 registered attempts satisfy the end-to-end mission-success criterion, with Candidate correctness, staged resolution, and mission completion observed throughout the campaign. The realized minimum inter-UAV distance remains above the registered $d_{\text{hard}} = 1.5$ m criterion in every successful execution, while the overall mean tracking RMSE is 0.085 m. This campaign connects the tested commitment, style-authority, and safety mechanisms across task boundaries, providing integration evidence over the evaluated missions and swarm sizes.

V. Conclusion

LATE-Swarm realizes under-specified natural-language UAV-swarm formation commands by preserving unresolved intent until the required state and planning information becomes available. Motion-style influence is bounded by predefined authority, while planning and feedback reduce predictable nominal and residual execution-time risks, respectively. Future work will investigate hardware experiments, richer dynamic missions, formally certified safety mechanisms, and transfer of the commitment principle to other multi-robot systems.

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